Derivatives: Contracts, Models, and Markets · Technical Note 27

The Exposure Profile of a Swap, Derived

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Figure 27.1 of Derivatives: Contracts, Models, and Markets draws three curves over the life of a five-year par swap—the expected exposure, the 95th-percentile potential future exposure, and the same exposure under daily variation margin with a ten-day margin period of risk—and Section 27.1 states the formulas behind them in a sentence. This note derives them. Under a driftless normal model for the swap rate the exposure at time t is the remaining annuity times the positive part of a normal variable, so its mean, its quantiles, the time at which it peaks, and its margined version all have closed forms; the hump sits at one third of the maturity for any volatility and any maturity; and the effective expected positive exposure that Section 27.5's multiplier α acts on is a one-line integral. The note closes with the two approximations the figure makes and what discounting and drift do to them.

1 SETUP

A payer swap on unit notional, entered at time 0 at the par rate $K = S(0)$, with maturity T and payment dates $T_1 < \cdots < T_n = T$. Write $A(t)$ for the annuity of the payments still to come at time t, the sum of the discount factors from t to each remaining date times its accrual fraction, and $S(t)$ for the forward swap rate of the remaining swap. Chapter 7's decomposition gives the swap's value to the payer as

$$V(t) = A(t)\,\big(S(t) - K\big), \tag{TN27.1}$$

which is zero at inception because $K = S(0)$.

The rate is modeled as normal with no drift and a constant normal volatility σ, quoted in rate units per square-root year:

$$S(t) = S(0) + \sigma\, W(t), \qquad W(t) \sim N(0, t), \tag{TN27.2}$$

so that $S(t) - K = \sigma W(t)$. The counterparty's failure matters only in the states where the swap is worth something to the surviving party, so the object of interest is the *exposure*

$$E(t) = \max\big(V(t), 0\big) = A(t)\,\sigma\, \max\big(W(t), 0\big) = A(t)\,\sigma\sqrt{t}\; Z^{+}, \tag{TN27.3}$$

where $Z = W(t)/\sqrt{t}$ is standard normal and $Z^{+} = \max(Z, 0)$. Everything that follows is a statement about Z^{+} scaled by the deterministic factor $A(t)\,\sigma\sqrt{t}$.

2 THE TWO CURVES

Expected exposure. The mean of the positive part of a standard normal is

$$\mathbb{E}\big[Z^{+}\big] = \int_0^{\infty} z\,\phi(z)\,dz = \big[-\phi(z)\big]_0^{\infty} = \phi(0) = \frac{1}{\sqrt{2\pi}} = 0.3989, \tag{TN27.4}$$

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using $\phi'(z) = -z\phi(z)$. Hence

$$\mathrm{EE}(t) = \mathbb{E}\big[E(t)\big] = \frac{A(t)\,\sigma\sqrt{t}}{\sqrt{2\pi}}. \tag{TN27.5}$$

Potential future exposure. For $x \geq 0$,

$$\mathbb{P}\big(E(t) \leq x\big) = \mathbb{P}\Big(Z \leq \frac{x}{A(t)\,\sigma\sqrt{t}}\Big) = N\Big(\frac{x}{A(t)\,\sigma\sqrt{t}}\Big),$$

so for any confidence level $q > 1/2$ the q-quantile of the exposure is

$$\mathrm{PFE}_q(t) = A(t)\,\sigma\sqrt{t}\;N^{-1}(q), \qquad \mathrm{PFE}_{0.95}(t) = 1.645\,A(t)\,\sigma\sqrt{t}. \tag{TN27.6}$$

(For $q \leq 1/2$ the quantile is zero: with probability one half the swap is against the surviving party and the exposure is nil.) The two curves therefore have the same shape and differ by the constant factor $N^{-1}(q)\sqrt{2\pi}$, which is 4.12 at the 95th percentile.

3 WHERE THE HUMP IS

Both curves are proportional to $A(t)\sqrt{t}$: the square root grows because the rate has had longer to travel, the annuity shrinks because fewer payments remain. Ignore discounting, so that the annuity is the remaining tenor, $A(t) = T - t$. Then

$$f(t) = (T-t)\sqrt{t}, \qquad f'(t) = -\sqrt{t} + \frac{T-t}{2\sqrt{t}} = \frac{T-3t}{2\sqrt{t}}, \tag{TN27.7}$$

which vanishes at

$$t^* = \frac{T}{3}, \qquad f(t^*) = \frac{2T}{3}\sqrt{\frac{T}{3}}, \tag{TN27.8}$$

and $f'' < 0$ there, so it is the maximum. The peak sits one third of the way through the swap's life whatever the volatility and whatever the maturity; the volatility scales the height and does not move it. For the figure's five-year swap, $t^* = 1.67$ years and $f(t^*) = 4.30$, so on \$100 million at $\sigma = 1\%$ the 95th-percentile PFE peaks at $100 \times 0.01 \times 4.30 \times 1.645 = \$7.08$ million and the expected exposure at $100 \times 0.01 \times 4.30 \times 0.3989 = \$1.72$ million—the heights the figure shows.

4 THE MARGINED CURVE

Under a daily variation margin agreement the mark-to-market is settled each day, so at any date the collateral held equals the value of the swap and the current exposure is nil. What is at risk is the future: if the counterparty fails at t, the surviving party is unhedged until the position is replaced, a window of length δ—the margin period of risk, ten business days or $\delta = 10/250$ of a year in the regulatory convention—and the loss is whatever the swap moves by during it. The margined exposure at t is therefore a forward-looking quantity, the value change over the window that a default at t opens. Over a window that short the annuity barely moves, so

$$E_\delta(t) = \max\big(V(t+\delta) - V(t), 0\big) \approx A(t)\,\sigma\,\max\big(W(t+\delta) - W(t), 0\big) = A(t)\,\sigma\sqrt{\delta}\;Z^+, \tag{TN27.9}$$

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since a Brownian increment over δ is $N(0,\delta)$ whatever t is. This is why the margined curve is positive at $t = 0$, when the swap is worth nothing and no margin has been posted: a counterparty that fails on the first day still leaves ten days of rate moves on a full-tenor swap uncovered, and that is the exposure the curve records. The margined curves are therefore (TN27.5) and (TN27.6) with $\sqrt{t}$ replaced by $\sqrt{\delta}$:

$$\mathrm{PFE}_q^{\delta}(t) = A(t)\,\sigma\sqrt{\delta}\;N^{-1}(q), \qquad \mathrm{EE}^{\delta}(t) = \frac{A(t)\,\sigma\sqrt{\delta}}{\sqrt{2\pi}}. \tag{TN27.10}$$

The square root of elapsed time has gone, and with it the hump: the margined exposure is largest at inception, when the most payments remain, and declines linearly with the tenor. At $t = 0$ on the figure's swap it is $100 \times 0.01 \times 5 \times 0.2 \times 1.645 = \$1.65$ million, against the unmargined peak of \$7.08 million.

The ratio of margined to unmargined exposure at the same date is $\sqrt{\delta/t}$, which depends on t. The standardized approach of Section 27.3 replaces that ratio with a single maturity factor, $1.5\sqrt{\delta}$ against an unmargined horizon of one year: 0.30 for ten days, where the model's ratio at $t = 1$ would be $\sqrt{10/250} = 0.20$. The 1.5 is the reciprocal of a $2/3$: the supervisory factors are calibrated to the one-year average of an exposure growing as $\sqrt{t}$, which is $\int_0^1 \sqrt{t}\,dt = 2/3$ of its one-year value, while the margined exposure is flat over its window and averages to its full value. Dividing by $2/3$ puts the two on one scale, which is the reading Section 27.3 gives it.

5 THE LOAN EQUIVALENT

Section 27.5's multiplier α acts on the *effective expected positive exposure*: the running maximum of the expected exposure, averaged over the first year,

$$\mathrm{EEPE} = \int_0^1 \max_{s \le t} \mathrm{EE}(s)\,dt, \tag{TN27.11}$$

For the figure's swap $\mathrm{EE}(t)$ is increasing on $[0,1]$, because its peak is at $t^* = 1.67 > 1$, so the running maximum is $\mathrm{EE}(t)$ itself and, with $A(t) = T - t$,

$$\mathrm{EEPE} = \frac{\sigma}{\sqrt{2\pi}}\int_0^1 (T-t)\sqrt{t}\,dt = \frac{\sigma}{\sqrt{2\pi}}\left[\frac{2}{3}T - \frac{2}{5}\right] = \frac{0.01}{\sqrt{2\pi}} \times 2.933 = 0.0117 \tag{TN27.12}$$

per unit of notional, or \$1.17 million on \$100 million. It is a loan-equivalent amount: the counterparty is treated as if it had borrowed \$1.17 million, and $\alpha = 1.4$ then lifts that to \$1.64 million for the reasons Section 27.5 gives. The figure is not comparable with Example 27.2's SA-CCR exposure of \$3.10 million, which comes from a supervisory factor rather than from a chosen volatility; the point of computing it is to show what the internal-model quantity that α multiplies actually is.

6 THE MODEL BEHIND SA-CCR

Section 27.3's standardized approach is this note's model made into arithmetic. The Basel Committee's account of its foundations takes the netting set's value over a one-year horizon to be normal with no drift, as in (TN27.2), and targets the EEPE of (TN27.11), which α then scales as it would a bank's own model output. Four correspondences follow.

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The supervisory factor. For a unit position whose value is $\sigma W(t)$, the one-year EEPE is $\int_0^1 \sigma\sqrt{t}\,\phi(0)\,dt = \frac{2}{3}\sigma/\sqrt{2\pi}$, and the supervisory factor is this number at a stressed normal volatility σ. Inverting, the 0.50% for interest rates corresponds to

$$\sigma = 0.005 \times 1.5 \times \sqrt{2\pi} = 1.88\% \tag{TN27.13}$$

a year in rate terms—188 basis points of normal volatility, a stressed level—applied to an effective notional that the supervisory duration has already converted into a rate sensitivity.

The maturity factor. The unmargined $\sqrt{\min(M, 1)}$ and the margined $1.5\sqrt{\delta}$ are the two averages of Section 4: an exposure growing as $\sqrt{t}$ averages to two-thirds of its one-year value, a margined exposure is flat over its window, and 1.5 is the ratio.

The aggregation. Within a hedging set the three maturity buckets are taken to be jointly normal, with correlation 0.7 between adjacent buckets and 0.3 between the first and the third, so the standard deviation of their sum is

$$\sqrt{D_1^2 + D_2^2 + D_3^2 + 2 \times 0.7\, D_1 D_2 + 2 \times 0.7\, D_2 D_3 + 2 \times 0.3\, D_1 D_3}, \tag{TN27.14}$$

which is the chapter's equation (27.8).

The multiplier. Let $m = V - C$ be the current uncollateralized value and $X \sim N(0, s^2)$ the change over the horizon, with s set so that $\mathbb{E}[X^+] = s\,\phi(0)$ equals the add-on. The exact ratio of the exposure of a position starting at m to that of one starting at zero is

$$\frac{\mathbb{E}[(m + X)^+]}{\mathbb{E}[X^+]} = \frac{m\,N(m/s) + s\,\phi(m/s)}{s\,\phi(0)}, \tag{TN27.15}$$

which equals one at $m = 0$ and has slope $N(0)/(s\,\phi(0)) = 1/(2\,\text{AddOn})$ there. The supervisory form $0.05 + 0.95\exp[m/(2 \times 0.95 \times \text{AddOn})]$ has the same value and the same slope at $m = 0$—it is the tangent exponential—and a floor of 5% that the exact ratio lacks. For Example 27.2's swap \$5 million out of the money, $s = 2.21/\phi(0) = 5.54$ and (TN27.15) gives 0.251 against the supervisory 0.339: the approximation is conservative, as a standardized formula is meant to be.

7 WHAT THE FIGURE ASSUMES

Two approximations separate Figure 27.1 from a full calculation, and both are easy to remove.

Discounting. With a flat continuously compounded rate r the annuity of the remaining tenor is $A(t) = (1 - e^{-r(T-t)})/r$ rather than $T - t$. The curves keep their form, $A(t)\,\sigma\sqrt{t}$ times a constant, and the first-order condition becomes $A(t) = 2t\,A'(t) \cdot (-1)$, that is, $1 - e^{-r(T-t)} = 2rt\,e^{-r(T-t)}$; at $r = 5\%$ and $T = 5$ the peak moves later, from 1.67 to 1.76 years, because discounting shrinks the far payments and so slows the annuity's decline, and its height falls by about eight percent. The one-third rule is the $r \to 0$ limit.

Drift. Capital is computed under a real-world distribution, and a swap rate with drift μ has $S(t) - K = \mu t + \sigma W(t)$, so the exposure is $A(t)\,\sigma\sqrt{t}\,\max(Z + \mu t/(\sigma\sqrt{t}), 0)$.

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The expected exposure becomes $A(t)\sigma\sqrt{t}\,[\phi(d) + d\,N(d)]$ with $d = \mu\sqrt{t}/\sigma$, which is the Bachelier call formula of Chapter 36 applied to a call on the rate struck at K—the exposure of a swap *is* a swaption payoff observed at t, which is why the chapter can say that the exposure is an option on the market. With $\mu = 0$, $\phi(0) + 0 = 1/\sqrt{2\pi}$ and (TN27.5) is recovered. The figure takes $\mu = 0$ because the point it makes is about the shape of the profile, and the drift moves the whole curve up or down without changing where it peaks by much.

Neither approximation touches the margined curve's derivation, which depends only on the increment over the window, nor the ratio $\sqrt{\delta/t}$ that Section 27.3 turns into the regulatory maturity factor.

REFERENCES

The exposure profile of a swap under a normal rate is worked in Gregory's *The xVA Challenge* (4th ed., 2020), whose treatment of expected exposure, potential future exposure, and effective EPE this note follows, and the Basel Committee's *The Standardised Approach for Measuring Counterparty Credit Risk Exposures* (2014) is the source of the maturity factor and of the ten-day and one-year horizons it compares, and the Committee's working paper *Foundations of the Standardised Approach for Measuring Counterparty Credit Risk Exposures* (2014) is where the normal model, the aggregation correlations, and the multiplier's exponential approximation are set out. Pykhtin and Zhu's 2007 guide to counterparty credit risk in *GARP Risk Review* is the short account of why the hump sits where it does and of the loan-equivalent construction that α corrects.

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